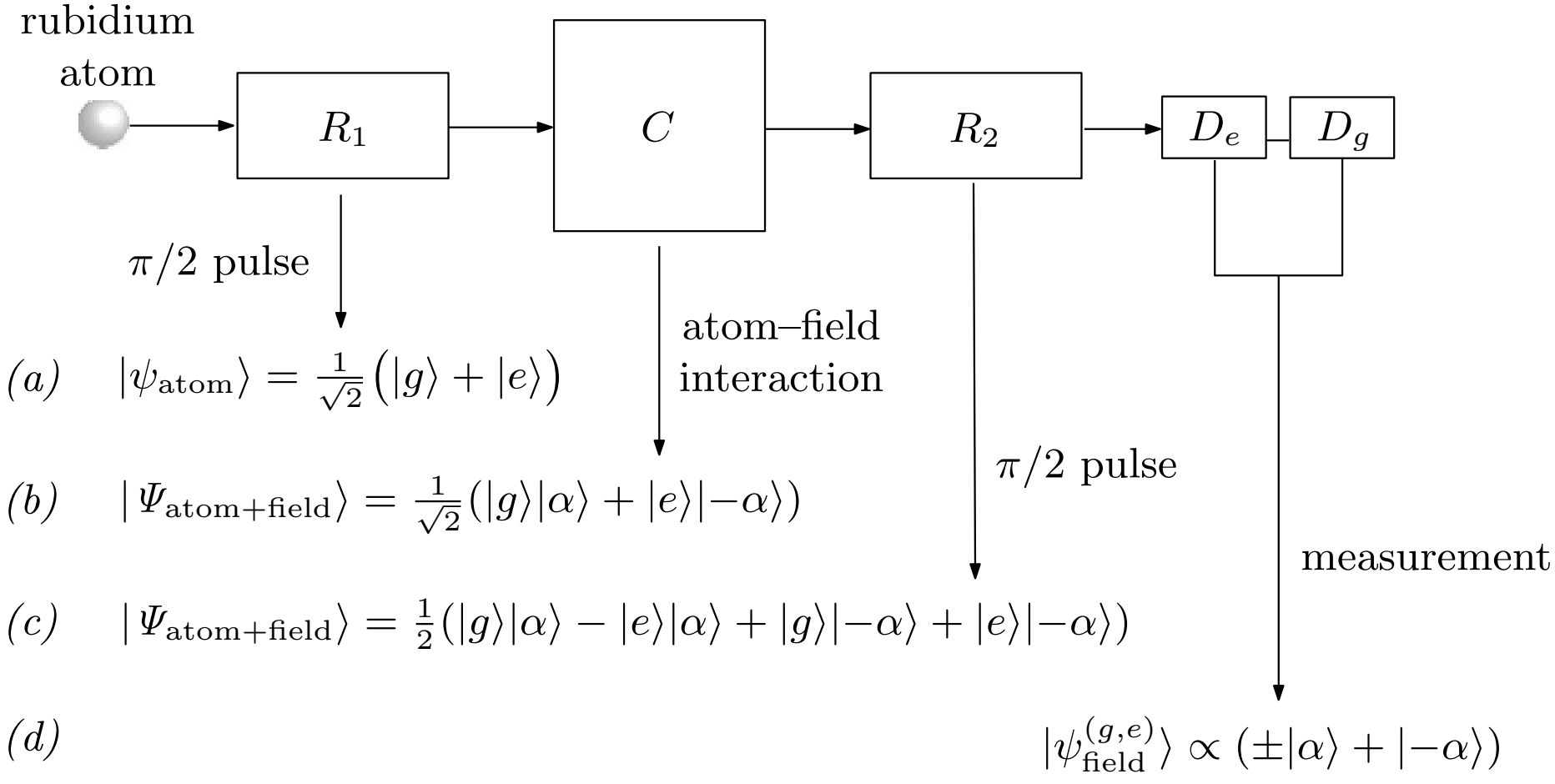

rubidium
atom
R_1
C
R_2
D_e
D_g
$\pi/2$ pulse
atom–field
interaction
$\pi/2$ pulse
measurement
(a) $|\psi_{\rm atom}\rangle = \frac{1}{\sqrt{2}}\big(|g\rangle + |e\rangle\big)$
(b) $|\Psi_{\rm atom+field}\rangle = \frac{1}{\sqrt{2}}(|g\rangle|\alpha\rangle + |e\rangle|-\alpha\rangle)$
(c) $|\Psi_{\rm atom+field}\rangle = \frac{1}{2}(|g\rangle|\alpha\rangle - |e\rangle|\alpha\rangle + |g\rangle|-\alpha\rangle + |e\rangle|-\alpha\rangle)$
(d) $|\psi^{(g,e)}_{\rm field}\rangle \propto (\pm|\alpha\rangle + |-\alpha\rangle)$